

413/613 Fall 2026 - Applied Statistics & Data Science I - Lecture 6

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RECALL

For a continuous random variable X with probability density $f_X(x)$, its cumulative distribution function (CDF) is

$$F_X(x) = \mathbb{P}(X \leq x) = \int_{-\infty}^x f_X(u) du, \quad \boxed{f_X(x) = \frac{d}{dx} F_X(x)}.$$

The CDF measures the **area under the PDF to the left of x** . Today we use this relation to find the distribution of a *function* of a random variable.

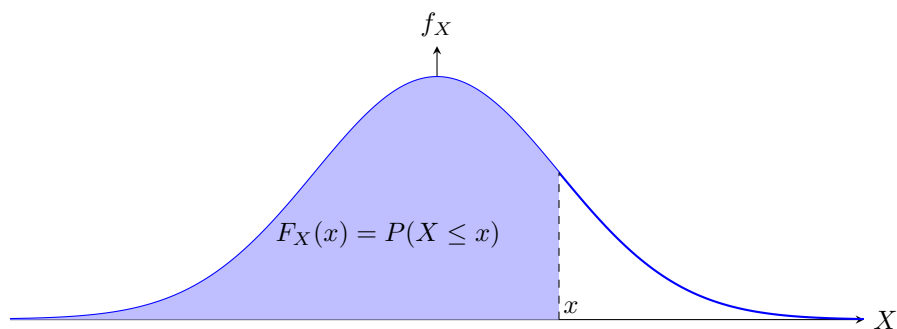


Figure 1: The CDF $F_X(x)$ is the shaded area under the PDF $f_X(x)$ to the left of x .

1 Standardization and the Normal Distribution

1.1 Why Standardize Random Variables?

Suppose Alice and Bob take two different exams. Who performed better relative to their class?

Student	Exam score x	Class mean μ	Class SD σ	z -score
Alice	85	70	10	1.5
Bob	42	30	5	2.4

Although Alice has a higher raw score, Bob performed better relative to his class. Standardization centers a random variable at zero and rescales it to unit variance, allowing quantities measured on different scales to be compared. Suppose X has mean μ and nonzero standard deviation σ . Define

$$\boxed{Z = \frac{X - \mu}{\sigma}}.$$

This transformation subtracts the center and measures deviations in units of the standard deviation. It gives

$$\mathbb{E}[Z] = 0, \quad \text{Var}(Z) = 1.$$

For an observed value x , the corresponding **z -score** is

$$\boxed{z = \frac{x - \mu}{\sigma}}.$$

It measures how many standard deviations an observation lies above or below the mean. For instance, $z = 2$ means two standard deviations above the mean, and $z = -1$ means one standard deviation below the mean.

In the above example with exam scores, the standardized score of Alice is

$$z = \frac{x - \mu}{\sigma} = \frac{85 - 70}{10} = 1.5$$

and of Bob is

$$z = \frac{x - \mu}{\sigma} = \frac{42 - 30}{5} = 2.4$$

Thus, Alice scored 1.5 standard deviations above her class mean, whereas Bob scored 2.4 standard deviations above his.

IMPORTANT: Standardization does *not* generally make a distribution Gaussian.

1.2 The Normal Distribution and Standardization

Suppose a random variable X follows a Gaussian (normal) distribution with mean μ and variance σ^2 :

$$X \sim \mathcal{N}(\mu, \sigma^2), \quad f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x - \mu)^2}{2\sigma^2}\right].$$

Recall that standardization consists of subtracting the mean and dividing by the standard deviation. For a Gaussian random variable, this linear transformation gives

$$X \sim \mathcal{N}(\mu, \sigma^2) \implies Z = \frac{X - \mu}{\sigma} \sim \mathcal{N}(0, 1).$$

The transformed random variable Z has mean zero, variance one, and follows the **standard normal distribution**. Its probability density function is denoted by $\phi(z)$:

$$\phi(z) = f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}.$$

The standard normal density is symmetric about zero, and the total area under its curve equals one. The parameters μ and σ determine the center and width of a general Gaussian distribution. Standardization removes these differences, allowing us to use the **same reference distribution** $\mathcal{N}(0, 1)$ for every Gaussian random variable.

Why is standardization useful? Suppose we want to calculate $P(X \leq x)$ for $X \sim \mathcal{N}(\mu, \sigma^2)$. By standardizing,

$$P(X \leq x) = P\left(Z \leq \frac{x - \mu}{\sigma}\right) = \Phi\left(\frac{x - \mu}{\sigma}\right),$$

where Φ is the cumulative distribution function of the standard normal distribution. Thus, probabilities for *any* Gaussian distribution can be calculated using a single reference CDF.

Connection to machine learning. Standardization is also widely used when different input features have different numerical scales. For example, age may be measured in years, while income is measured in thousands of dollars. A common preprocessing step is to standardize each feature using its training-set mean and standard deviation:

$$z_j = \frac{x_j - \hat{\mu}_j}{\hat{\sigma}_j}.$$

This produces dimensionless features on comparable scales and can improve numerical conditioning during optimization. Related centering and rescaling operations also appear in neural-network normalization layers.

1.3 The Standard Normal CDF and Survival Function

For $Z \sim \mathcal{N}(0, 1)$, define the **standard normal CDF**

$$\Phi(z) = P(Z \leq z) = \int_{-\infty}^z \phi(u) du.$$

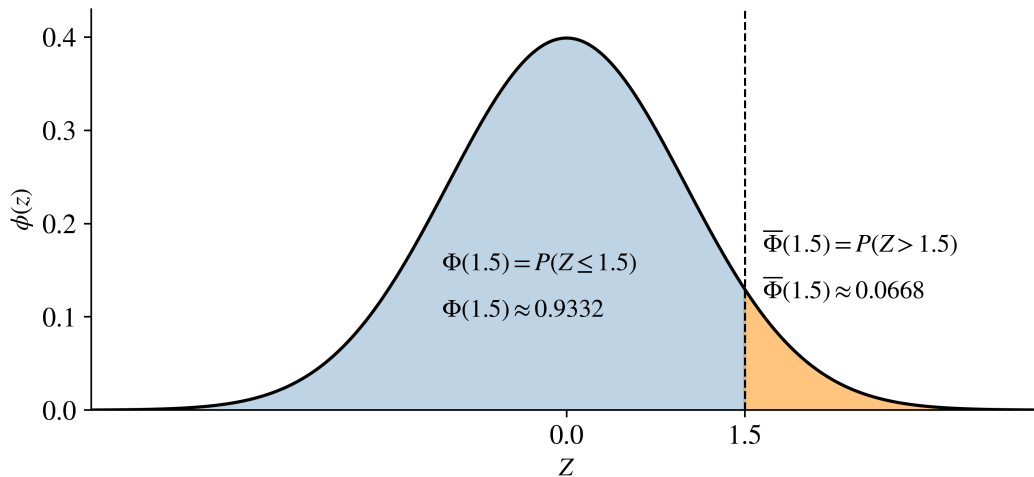


Figure 2: **Standard normal CDF and survival function.** For $Z \sim \mathcal{N}(0,1)$, the blue shaded area represents the cumulative distribution function $\Phi(1.5) = P(Z \leq 1.5) \approx 0.9332$. The orange shaded area represents the complementary probability, or survival function, $\bar{\Phi}(1.5) = P(Z > 1.5) = 1 - \Phi(1.5) \approx 0.0668$. Together, the two shaded regions account $(\Phi(1.5) + \bar{\Phi}(1.5)=1)$ for the total probability of one under the standard normal density $\phi(z)$.

Geometrically, $\Phi(z)$ is the **area under the standard normal curve to the left of z** (See Figure). The integral above does not have a closed-form expression in terms of elementary functions. Its values are therefore tabulated in a **standard normal (Z) table**. For example, using a table that reports $\Phi(z) = P(Z \leq z)$, we find

$$\Phi(1.50) \approx 0.9332.$$

Hence,

$$P(Z > 1.50) = 1 - \Phi(1.50) \approx 0.0668.$$

The complementary area to the right is the **survival function** (upper-tail probability):

$$\boxed{\bar{\Phi}(z) = P(Z > z) = 1 - \Phi(z).}$$

For any $a < b$, the area between a and b is

$$P(a \leq Z \leq b) = \Phi(b) - \Phi(a).$$

By symmetry of the standard normal distribution, for negative values, we use the relation

$$\Phi(-z) = 1 - \Phi(z).$$

Example.

$$P(-1.96 \leq Z \leq 1.96) = \Phi(1.96) - \Phi(-1.96) = 2\Phi(1.96) - 1 \approx 0.95.$$

For any Gaussian random variable $X \sim \mathcal{N}(\mu, \sigma^2)$, standardization gives

$$F_X(x) = P(X \leq x) = P\left(Z \leq \frac{x - \mu}{\sigma}\right) = \Phi\left(\frac{x - \mu}{\sigma}\right).$$

Thus, probabilities for *any* Gaussian distribution can be computed using a single standard normal CDF Φ , whose values are available from numerical methods or standard normal tables.

Example. If $X \sim \mathcal{N}(100, 15^2)$, then

$$\begin{aligned} P(X \leq 130) &= P\left(Z \leq \frac{130 - 100}{15}\right) \\ &= \Phi(2) \approx 0.9772. \end{aligned}$$

Thus, approximately 97.7% of the probability lies to the left of 130.

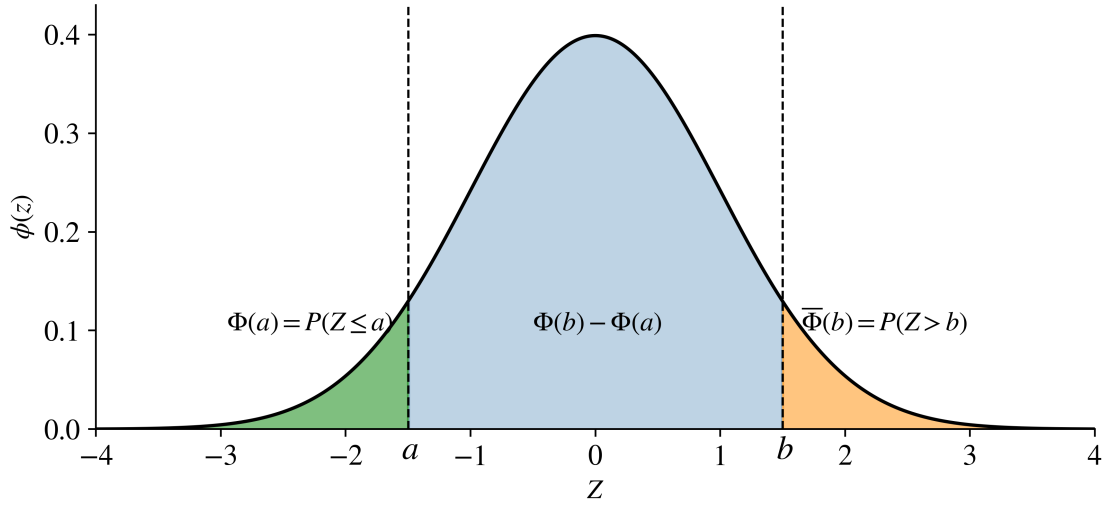


Figure 3: **Probabilities under the standard normal distribution.** For $Z \sim \mathcal{N}(0, 1)$ and $a < b$, the green shaded area represents the cumulative probability $\Phi(a) = P(Z \leq a)$. The blue shaded area represents the probability between a and b , $P(a \leq Z \leq b) = \Phi(b) - \Phi(a)$. The orange shaded area represents the survival probability $\bar{\Phi}(b) = P(Z > b) = 1 - \Phi(b)$. The three regions partition the total area under the standard normal density $\phi(z)$, and their probabilities sum to one.

2 Functions of Random Variables

Suppose we know the distribution of random variable X , but the quantity of interest is a new random variable

$$Y = g(X).$$

where $g(\cdot)$ is some function of X .

QUESTION: If X has PDF $f_X(x)$, what is the PDF $f_Y(y)$ of the transformed variable? Our approach is to compute the CDF of Y first, and then differentiate:

$$F_Y(y) = P(Y \leq y) = P(g(X) \leq y); \quad f_Y(y) = \frac{d}{dy} F_Y(y).$$

The key step is to express the event $g(X) \leq y$ as an event involving X , whose distribution we already know.

2.1 Example 1: Linear Transformation $Y = aX + b$

Let $a > 0$ and define $Y = aX + b$. Then

$$\begin{aligned} F_Y(y) &= P(aX + b \leq y) \\ &= P\left(X \leq \frac{y-b}{a}\right) \\ &= F_X\left(\frac{y-b}{a}\right). \end{aligned}$$

Differentiating with respect to y and applying the chain rule,

$$\begin{aligned} f_Y(y) &= \frac{d}{dy} F_X\left(\frac{y-b}{a}\right) \\ &= f_X\left(\frac{y-b}{a}\right) \frac{1}{a}. \end{aligned}$$

$$f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right), \quad a \neq 0. \quad (1)$$

The absolute value makes the formula valid for $a < 0$ as well (in that case, the inequality reverses when computing the CDF). The factor $1/|a|$ ensures that the transformed PDF still integrates to one.

The mean and variance transform as

$$\mathbb{E}[Y] = a\mu + b, \quad \text{Var}(Y) = a^2\sigma^2,$$

where $\mu = \mathbb{E}[X]$ and $\sigma^2 = \text{Var}(X)$.

For the case when $X \sim \mathcal{N}(\mu, \sigma^2)$ and define $Y = aX + b$, where $a \neq 0$. The PDF of X is

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right].$$

Using the change-of-variables formula in Eq. (1),

$$\begin{aligned} f_Y(y) &= \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right) \\ &= \frac{1}{|a|\sqrt{2\pi\sigma^2}} \exp\left[-\frac{\left(\frac{y-b}{a} - \mu\right)^2}{2\sigma^2}\right] \\ &= \frac{1}{\sqrt{2\pi a^2\sigma^2}} \exp\left[-\frac{(y - (a\mu + b))^2}{2a^2\sigma^2}\right]. \end{aligned}$$

Comparing with the Gaussian PDF, we identify

$$Y \sim \mathcal{N}(a\mu + b, a^2\sigma^2).$$

Conclusion: A linear transformation of a Gaussian random variable remains Gaussian, with

$$\mathbb{E}[Y] = a\mu + b, \quad \text{Var}(Y) = a^2\sigma^2.$$

2.2 Example 2: Non-linear transformation $Y = X^2$

We now consider a non-linear transformation $Y = X^2$. To derive the form of $f_Y(y)$, first take the simpler case $X \sim \mathcal{N}(0, 1)$.

We follow a similar procedure as above and first derive the form of CDF $F_Y(y)$,

$$\begin{aligned} F_Y(y) &= P(Y \leq y) \\ &= P(X^2 \leq y) \\ &= P(-\sqrt{y} \leq X \leq \sqrt{y}) \\ &= F_X(\sqrt{y}) - F_X(-\sqrt{y}) \\ &= F_X(\sqrt{y}) - (1 - F_X(\sqrt{y})) \\ &= 2F_X(\sqrt{y}) - 1 \end{aligned}$$

Differentiating (for $y > 0$),

$$f_Y(y) = 2f_X(\sqrt{y}) \frac{1}{2\sqrt{y}} = f_X(\sqrt{y}) \frac{1}{\sqrt{y}}$$

For the standard normal PDF,

$$f_X(x) = \phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \quad \phi(-x) = \phi(x),$$

so

$$f_Y(y) = f_X(\sqrt{y}) \frac{1}{\sqrt{y}} = \frac{1}{\sqrt{2\pi y}} e^{-y/2}, \quad y > 0.$$

We call this the **Chi-square distribution with one degree of freedom**, $Y \sim \chi_1^2$. The important distinction from the linear case is that $x \mapsto x^2$ is *not one-to-one*: both $+\sqrt{y}$ and $-\sqrt{y}$ contribute to the PDF.

TAKEAWAY: To find the distribution of $Y = g(X)$, first compute $F_Y(y) = \mathbb{P}(g(X) \leq y)$ and then differentiate the CDF $F_Y(y)$ to derive the PDF. $f_Y(y)$. For Gaussian random variable X , standardization reduces all interval and tail probabilities to areas under a single standard normal curve, which in turn can be computed using Φ and $1 - \Phi$.